Module 1

1. In which OSI layer the Wi-Fi standards/protocol fits?

Wi-Fi (IEEE 802.11) operates in:

- Layer 1 : Physical Layer
  - Handles transmission of data using radio waves (2.4 GHz, 5 GHz, 6 GHz)
  - Defines modulation techniques like OFDM
- Layer 2 : Data Link Layer
  - Stores bit received from physical layer into frames
  - Uses MAC protocol (CSMA/CA) to control access
  - Handles framing, addressing (MAC address), and error detection

2. Can you share the Wi-Fi devices that you are using day to day life share that device's wireless capability / properties after connecting to the network. Match your device to Wi-FI generation based on the capability

1. Laptop

- SSID: Airtel_siva_3012
- Frequency Band: 5 GHz
- Protocol: Wi-Fi 5 (802.11ac)
- Security: WPA2-Personal
- Link Speed: 433 Mbps
- Wi-Fi Adapter: Realtek RTL8852BE

Wi-Fi Generation: Wi-Fi 5 (802.11ac)

2. Smartphone

- Frequency Band: 2.4 GHz
- Link Speed: 72 Mbps
- Security: WPA/WPA2-Personal

Current Wi-Fi Generation: Wi-Fi 4 (802.11n)
Device Capability: Supports Wi-Fi 5 (802.11ac)

3. Home Wi-Fi Router (Airtel Router)

- Frequency Band: 2.4 GHz and 5 GHz (Dual-band)
- Protocol: Wi-Fi 5 (802.11ac)
- Security: WPA2-Personal
- Supports multiple devices

Wi-Fi Generation: Wi-Fi 5 (802.11ac)

3. What is BSS and ESS ?

BSS (Basic Service Set)

- A Basic Service Set (BSS) is a Wi-Fi network formed by a single Access Point (AP) and the devices connected to it.
- It covers a small area and is identified by the BSSID, which is the MAC address of the access point.
- Example: A home Wi-Fi network with one router.

ESS (Extended Service Set)

- An Extended Service Set (ESS) is formed by connecting multiple BSSs together through a wired network.
- All access points share the same SSID, allowing users to move between them without losing connection.
- Example: Wi-Fi network in a college or office campus.

4. What are the basic functions of Wi-Fi access points ?

- Connects wired network to wireless devices (bridge between Ethernet and Wi-Fi)
- Provides wireless connectivity to devices like mobiles and laptops
- Acts as a central point for communication between connected devices
- Extends Wi-Fi coverage to weak signal areas (repeater function)
- Supports multiple networks (SSIDs) for users like guests or employees
- Can connect different network types (e.g., wireless devices to wired network)

5. Difference between Bridge Mode and Repeater Mode.

Bridge Mode	Repeater Mode
Connects two networks	Extends Wi-Fi coverage
Used between buildings or LANs	Used in homes for dead zones
No major speed loss	Speed reduces
Links two separate networks	Uses same network and repeats signal
Fully used for connection	Shared between original and repeated signal
More stable connection	Less stable compared to bridge

6. What are the difference between IEEE 802.11a and IEEE 802.11b

Feature	IEEE 802.11a	IEEE 802.11b

Frequency Band	5 GHz	2.4 GHz
Maximum Speed	54 Mbps	11 Mbps
Range	Shorter	Longer
Interference	Less interference	More interference
Modulation	OFDM	DSSS
Channels	More non-overlapping channels	Limited channels
Cost	Higher initially	Lower

7. Configure your modem/hotspot to operate only in 2.4 Ghz and connect your laptop / WI-Fi device and capture the properties in your wifi device. Repeat the same in 5Ghz and tabulate the difference you observe during this.

Feature	2.4 GHz	5 GHz
Frequency Band	2.4 GHz	5 GHz
Wi-Fi Standard	Wi-Fi 4 (802.11n)	Wi-Fi 5 (802.11ac)
Channel	11	149
Link Speed (Rx/Tx)	72 / 72 Mbps	433 / 433 Mbps
Security	WPA2-Personal	WPA2-Personal

8. What is the difference between IEEE and WFA ?

IEEE	WFA (Wi-Fi Alliance)
Institute of Electrical and Electronics Engineers	Wi-Fi Alliance
Standards organization	Industry consortium
Develops Wi-Fi standards	Certifies Wi-Fi devices

Defines protocols (802.11 series)	Ensures devices work together
Standards like 802.11a/b/g/n/ac/ax	“Wi-Fi Certified” label
Technical design	Testing and compatibility

9. List down the types of Wi-Fi internet connectivity backhauls share your home/college’s wireless internet connectivity backhaul name and its properties.

Types of Wi-Fi backhauls:

- Fiber optic
- Cable / MoCA
- Cellular (4G/5G)
- Satellite
- Powerline

Backhaul used in home:

Type: Fiber optic backhaul (Airtel Broadband)

- High speed internet
- Low latency
- Reliable connection
- Less interference
- Supports multiple users

10. List down the Wi-Fi topology and their use cases.

- **Infrastructure Mode**
  - Devices connect through a central access point (router).
  - Use case: Home, office, and campus networks for regular internet access.
- **Repeater Mode**
  - Extends the existing Wi-Fi signal to cover weak or dead zones.
  - Use case: Large homes or buildings with poor signal areas.
- **Bridge Mode**
  - Connects two separate networks wirelessly.
  - Use case: Connecting two buildings without using cables.
- **Ad-Hoc Mode**
  - Devices connect directly to each other without an access point.
  - Use case: File sharing or temporary connections between devices.